

Advanced (Carolina Reaper)

Q1. Solve the inequality $2x + 5 > 11$.

- (A) $x > 3$
- (B) $x < 3$
- (C) $x \geq 3$
- (D) $x \leq 3$
- (E) $x = 3$

Q2. Graph the solution to $x - 2 < 7$ on a number line.

- (A) $(-\infty, 9)$
- (B) $(-\infty, 5)$
- (C) $(-9, \infty)$
- (D) $(-5, \infty)$
- (E) $(-7, \infty)$

Q3. Write the solution to $x > 4$ and $x < 10$ in interval notation.

- (A) $(4, 10)$
- (B) $[4, 10]$
- (C) $(-\infty, 4) \cup (10, \infty)$
- (D) $(-\infty, 10] \cup [4, \infty)$
- (E) $[4, 10)$

Q4. Solve the inequality $-3x + 2 < 14$.

- (A) $x > -4$
- (B) $x < -4$
- (C) $x \geq -4$
- (D) $x \leq -4$
- (E) $x = -4$

Q5. Solve the inequality $\frac{x}{2} + 1 > 3$.

- (A) $x > 4$
- (B) $x < 4$
- (C) $x \geq 4$
- (D) $x \leq 4$
- (E) $x = 4$

Q6. Graph the solution to $x + 1 > 5$ on a number line.

- (A) $(-\infty, 4)$
- (B) $(-\infty, 6)$
- (C) $(-4, \infty)$
- (D) $(-6, \infty)$
- (E) $(-1, \infty)$

Q7. Write the solution to $x - 3 > 2$ in interval notation.

- (A) $(5, \infty)$
- (B) $(-\infty, 5)$
- (C) $[5, \infty)$
- (D) $(-\infty, 5]$
- (E) $(-5, \infty)$

Q8. Solve the inequality $-2x - 5 < 1$.

- (A) $x > -3$
- (B) $x < -3$
- (C) $x \geq -3$
- (D) $x \leq -3$
- (E) $x = -3$

Open Ended Questions (FRQ)

Q9. Solve the inequality $x^2 - 4x - 3 > 0$ and graph the solution on a number line.

Q10. Write the solution to the system of inequalities $x + y > 2$ and $x - y < 1$ in interval notation.

Chef's Tips

- Review the concepts of inequalities before attempting the test
- Make sure to read each question carefully and choose the correct answer
- Use a number line to help visualize the solutions to the inequalities